



Programme: RDS (Group: 4)

Assignment

BAIT1093 INTRODUCTION TO COMPUTER SECURITY ASSIGNMENT (052024)

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Date of Submission: 6 September 2024

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Part 1: Understanding and Application of Security Measures

Criteria	Unsatisfactory (0 marks)	Satisfactory (1-2 marks)	Good (3-4 marks)	Excellent (5-6 marks)	Total
Introduction	Fails to provide an introduction or the introduction is irrelevant.	Provides a basic overview of the topic with minimal detail on its importance.	Provides a clear overview of the topic and its relevance.	Provides a comprehensive overview of the topic, clearly outlining its significance in computer security.	
Application of Concepts	Fails to demonstrate understanding of the topic's application in security.	Shows a basic understanding of the topic's application with limited examples and analysis.	Shows a good understanding of the topic's application with clear examples and analysis.	Demonstrates an excellent understanding of the topic's application in protecting systems, with detailed examples and in-depth analysis.	
Case Studies	Does not include case studies or the case studies provided are irrelevant.	Includes basic case studies with minimal analysis of outcomes and lessons.	Includes case studies with clear analysis of outcomes and lessons, but less detail.	Includes detailed case studies showing the real-world application of security measures, with insightful analysis of outcomes and lessons learned.	
Challenges and Solutions	Fails to discuss challenges or solutions related to the security measures.	Briefly mentions challenges with basic or generic solutions.	Discusses challenges with some proposed solutions, but lacks depth or evidence.	Provides an in-depth analysis of challenges and proposes innovative solutions or improvements, supported by evidence.	

Conclusion and Future Directions	Fails to provide a conclusion or suggestions for future directions.	Provides a basic conclusion with limited or generic suggestions for future directions.	Provides a clear conclusion and suggestions for future directions, with some insight.	Offers insightful conclusions and suggests future research or implementation directions, demonstrating a deep understanding of the topic.	
Total					

Part 2: Analysis of Computer Security Technologies

Criteria	Unsatisfactory (0 marks)	Satisfactory (1-2 marks)	Good (3-4 marks)	Excellent (5-6 marks)	Total
Overview	Fails to recap the topic or the overview is irrelevant.	Offers a basic overview of the topic with limited insight into its significance.	Provides a clear recap of the topic and its significance.	Concise recap of the topic with an emphasis on its broader significance in computer security.	
Threat Analysis	Fails to analyze the security attacks and threats that the technology addressed.	Gives a basic analysis with limited examples and evidence.	Offers a clear analysis with examples, but may lack some detail or depth.	Provides a thorough analysis of security attacks and threats that the technology addressed, supported by detailed examples and evidence.	
Evaluation of Effectiveness	Fails to evaluate the technology's effectiveness or assess vulnerabilities.	Provides a basic assessment of effectiveness with minimal critical insight.	Assesses the technology's effectiveness with some critical insights, but lacks depth.	Critically assesses the technology's effectiveness, including vulnerabilities, with a deep understanding of implications.	
Future Directions	Fails to discuss future directions or the discussion is irrelevant.	Mentions future trends and challenges with limited or generic suggestions.	Discusses future trends and challenges with some suggestions for advancement.	Provides insightful analysis of emerging trends and challenges, with suggestions for advancing the field.	
Conclusion	Fails to provide a conclusion or the conclusion is irrelevant.	Provides a basic conclusion with limited final thoughts.	Provides a clear conclusion with final thoughts, but may lack depth.	Offers insightful conclusions reflecting on the importance of computer security technologies, with clear final thoughts.	

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1.0 Introduction

Biometric Authentication is the cutting edge technology that uses our unique biological characteristics to verify our identity. This technology can be traced back in ancient times when simple fingerprints and handprints were used as signatures and seals. Based on the (*Biometry Authentication History (Infographic)*, 2023), the tool of identification such as biometrics began in the late 19th century with the work of a French police officer, Alphonse Bertillon. He is a biometrics researcher who applied the anthropological technique of anthropometry to law enforcement creating an identification system based on physical measurements such as height, weight, and the size of the head, arms, and feet, to identify criminals. This system soon after was considered as the golden standard for identification until the early 20th century which is when the first fingerprinting system was developed. In the latter half of the 20th century, biometric systems such as voice, iris and facial recognition were developed and became commercially viable in the 2010s and widely used since then.

In terms of how it works, the system will capture and store the biometric data of a user like a record into a secure database. The data will be converted into a template which will be used to compare with the user's biometric data when the user attempts to gain access to a system. Access will be granted when the match is sufficiently close.

It is important to note that biometric authentication is tremendously significant in safeguarding the computer system. One of the reasons being that it enhances security in a big way. The major selling point of biometric is that all of its traits are unique to each individual which will make it difficult to duplicate or steal. This would replace the traditional password-based system because it does not have the same vulnerabilities such as password cracking, phishing and social engineering. These risks could be reduced because it does not rely on something that can be easily forgotten, guessed or stolen by anyone. Having that being said, users will have a better experience because of it and insider threats no longer pose a threat as it used to be.

1.1 Application

Nowadays, different versions and types of biometric authentication systems have been implemented into our daily lives in order to protect important information due to the large demand in technology advancement and the recent increase in cyber crime throughout the years. Biometric authentication systems can be seen everywhere in our daily lives such as fingerprint recognition, facial recognition, iris recognition, voice recognition, DNA matching and so on.

Fingerprint Recognition

Based on (*About Touch ID Advanced Security Technology*, 2023), one of the few noticeable biometric authentication systems is fingerprint recognition. Every single human being has distinct patterns of ridges and valleys on their fingertips. With that being said, everyone will have different fingerprints which result in one of the best ways to identify someone's identity through scanning their fingerprint. In most cases, fingerprint biometric authentication can be used to manage access control.

The most common application of fingerprint authentication systems that we will see in our daily lives is on our smartphones. Smartphone manufacturers have fully adapted the fingerprint technology to tightly secure the access of the devices. With just a simple touch of our fingertips, this feature gives us a seamless and quick way to unlock our devices without the need of a password or a PIN number. Other than enhancing the user's experience and convenience, it also tightens the security of our devices and reduces the chances of unauthorised access to happen.

Majority of the banks in most countries have also applied fingerprint authentication to their system which improved the bank's security significantly. Any transactions to be made by using the ATMs or any online means must go through the fingerprint recognition process. Users have to have their fingerprint scanned in order to gain access to either withdraw, deposit their money or to make payments without the usage of a card or PIN number. This reduces the potential risk of card skimming and PIN theft.

Facial Recognition

Based on (*Face ID and Touch ID Security – Apple Support (UK)*, 2021), facial recognition is one of the newly added applications of biometric authentication that we encountered in our daily lives. This technology focuses on the facial features of a person to verify the identity of an individual. An individual's facial features such as the distance between the eyes, the shape of the

nose, the contours of the jawline and so on are all very unique and distinctive that it is proven to be a herculean task to replicate it. These facial features will be captured, analysed, and saved in the system and it will be used to compare with the facial data provided during subsequent authentication attempts.

A real life example would be our own devices. Due to the rapid growth of popularity in technologies, devices such as our smartphones have implemented facial recognition to manage the access of these devices and to authenticate the users of said devices. Our smartphones will capture and save a real-time image of our face into its database and it will be used when the users want to unlock their smartphones by comparing their facial features with the stored template. During the process, when the given image matches the stored template by a certain percentage, it will grant the user full access to the smartphone. This process happens almost instantaneously, providing both security and convenience to the user.

Another real life example that applies facial recognition is the airports. Airports around the world have been using facial recognition technology to streamline and secure the passenger experience. When passengers check in and pass through security checkpoints, the facial recognition system will scan and verify the passengers identity and match it with their passport photos. The system improves the security and the efficiency of the entire process.

Iris Recognition

Based on (*Iris Recognition: Biometric Authentication* | NEC, n.d.), iris recognition is a highly accurate biometric authentication that performs identity verification by analyzing and comparing the unique patterns of the iris in the human eye. The iris is the colored part of the eye surrounding the pupil. Its structure is unique to each person, and even twins have different iris patterns. The pattern of the iris remained essentially the same over the course of life, so the iris is a reliable long-term biometric feature that can be used as a way to identify an individual. With that being said, iris recognition has an extremely low error rate and is one of the most accurate biometric authentication technologies. The identification process is usually completed within seconds and is suitable for high-traffic environments.

For instance, Google uses iris recognition technology in its global data centres to ensure that only authorized employees have access to sensitive areas of the data centres. This is to prevent unauthorized personnel from accessing servers and storage devices and maintain data security.

Voice Recognition

Based on (Gillis & Mellon, n.d.), voice recognition uses the unique characteristics of the users' voice to verify users identity. This technology collects the user's voice samples and analyses various features of the voice such as the pitch, tone, cadence, and accent, to build an accurate voiceprint model which will be used for comparing purposes such as most of the biometric authentication technologies. After collecting the user's voice samples, machine learning algorithms will help to train acoustic models and allow the system to learn the user's voice print characteristics by analyzing a large amount of voice samples. The system will make a verification decision based on the matching results.

There are some banks and financial institutions that use voice recognition to verify customer identities to ensure account security. For instance, customers can use voice recognition to authenticate themselves when calling a bank's customer service number without having to enter a password.

1.2 Case Study

The rising growth of biometric authentication technology has garnered significant attention due to its increasing adoption across various sectors such as banking, healthcare, and personal security.

1. Apple's Face ID Biometric Authentication

Based on (*Face ID and Touch ID Security – Apple Support (UK)*, 2021), one of the examples would be our smartphones, specifically the iPhone made by Apple Inc. iPhone is a series of smartphones that are famous for its capabilities providing a combination of features and services such as mobile telephone, digital camera, music player, and personal computing technologies. The topic that is going to be the spotlight is Apple's biometric authentication technology. iPhone's Face ID are famous for their tremendous impact on both security and user convenience in serving as biometric authentication methods by using the facial feature data of an individual.

Apple's biometric security architecture heavily relies on a strict separation of responsibilities between the biometric sensor and the Secure Enclave which is a dedicated secure subsystem integrated into the system. The biometric sensor captures the biometric image of an individual's facial feature and securely transmits it to the Secure Enclave. The Secure Enclave will process, encrypt and store the corresponding Face ID template data. The stored templates will then be used to compare with incoming data from the biometric sensor to determine the authorisation of the user to unlock the device or for making payments such as in-app purchases.

Outcome and Impact

Since Apple introduced Face ID as the newest method to unlock the phone or make authenticate payments by using facial recognition, it has set a new standard for smartphone security by combining ease of use with high level of security. More devices and electronics have started to adopt this method to tighten their devices' security, increasing the expectation of consumers for the best protection against any unauthorised access of their devices in the future.

Insights Gained

Valuable important lessons have surfaced when implementing biometric authentication systems such as Face ID on an iPhone. One of them is balancing strong security with user's convenience is crucial as it provides robust security by using facial recognition. It is still being done in a way that is quick and easy for users. Another valuable lesson is that biometric systems need to adapt to various conditions, especially user's minor appearance changes. It is a known fact that anyone's appearance will change overtime and it will highly affect the process of biometric authentication. Additionally, it is important for the system to provide some sort of fallback mechanism in such a case where biometric authentication may not be feasible in order to maintain accessibility and functionality.

2. HSBC Bank's Voice ID Biometric Authentication

Another example is HSBC Bank's Voice ID biometric authentication system. HSBC Bank is one of the largest banks and one of the largest banking and finance services organisations in the world. Due to its massive growth and high expectations from users, HSBC Bank has developed and implemented voice ID biometric authentication for the purpose of enhancing security and streamline customer service. HSBC Bank uses the characteristics of a person's voice to verify their identity in order to allow account access to the users.

Based on (*HSBC Telebanking | Benefits and How to Get Started - HSBC MY*, n.d.), HSBC Bank's Voice ID creates the user's unique voiceprint by combining the physical factors such as vocal tract, mouth shape and size, nasal passage, and the behaviour factors such as speed of speech, pronunciation and emphasis, accents. The voiceprint will then be compared with incoming voice lines made by the users in order to verify their identities.

Outcome and Impact

With the new voice ID biometric authentication system being implemented into HSBC Banks, it has improved customers' convenience by removing the need of remembering passwords, pin numbers and security questions. Furthermore. It has also increased the difficulty for any frauds or cases of impersonation from occurring due to the fact that HSBC Bank has developed countermeasures such as they have measured more than 100 physical and behavioural factors of how a voice is produced, rather than the sounds themselves.

Insight Gained

One of the key factors for the success of the system is its effectiveness in combating fraud and impersonation. However, the system can still be vulnerable to new fraud techniques as time passes. It is highly acute and critical for HSBC Bank to constantly monitor and update their systems to reduce cases of fraud and impersonation. HSBC Bank has also made sure that the system is able to adapt to any variety of voices by making procedures such as implementing the under 18 restriction for the cause that young customers will have changes in the physical factors of their voices when growing up that could result in failed verification attempts which will highly impact the overall user experience and the security of the account.

3. Malaysia's MyKad Identification Card

Based on ((PDF) *Security Issues on Identity Card in Malaysia*, n.d.), MyKad is one of the few examples that applies biometric authentication systems. MyKad is Malaysia's pride and joy as it serves as the national identity card and also as transit cards through the service Touch 'n Go till this day. Originally, MyKad only contained 4 applications, which are the national identity card, driving licence, passport information, and health information. As time goes on and the speedy growth in biometric authentication technologies, new applications have been added to the identity card specifically for online transaction purposes. MyKad has become a core part of Malaysia's daily operation.

MyKad contains a smart chip to store biometric data such as fingerprints and it will be used as a template to compare incoming data obtained from scanning an individual's fingerprint, allowing access to users in using facilities or making online transactions.

Outcome and Impact

Due to its outstanding usage, it has already made an impact by reducing the need of traditional passwords or PINs, allowing quick verifications to be made. With biometric authentication, it has been proven that cases of fraud and impersonation have been reduced significantly, improving national security overall.

Insights Gained

First thing to be noted is that by implementing biometric authentication systems, it raises concern about data privacy and security. As such, proper measures need to be taken in place in order to protect biometric data from any unauthorised access or breaches.

MyKad's biometric authentication system must be always accurate and reliable to prevent cases of false rejection or acceptance from impacting user experience and security.

1.3 Challenges and Solutions

Biometric authentication system is a type of technology that uses biological characteristics to verify the identity of an individual, and it can also act as a second factor authentication. It is crucial because it provides a stronger security and lower the risk of identity theft of an individual by evaluating their biological traits. However, as mentioned above, implementing biometric authentication systems is not easy as there are several challenges in doing so. Even after the implementation of said systems, there are also a few things to be considered in order for the systems to proceed the tasks with pinpoint accuracy and maximum efficiency while also having security measures being taken.

1. Enrollment Failure

Firstly, one of the challenges of implementing a biometric authentication system is it may fail to enrol biometric information into the system. It occurs when a template for biometric information cannot be successfully created into a database or system.

A. Poor Quality of Biometric Data

There are several factors that may lead to failure in enrolling the biometric information. For example, the poor quality of biometric data due to dirt or a person who has a physical or medical condition such as absence or very less presence of fingerprint can lead to failure in enrolling their biometric data into the biometric authentication system. For instance, some individuals that have sweaty hands may encounter challenges because the moisture can smudge or distort the fingerprint which makes the biometric authentication system difficult to capture and recognise their fingerprints. This poses significant obstacles in tasks such as obtaining identification cards or accessing a bank account which both need fingerprint recognition as a biometric authentication. As a result, individuals with these conditions may find it difficult to complete essential processes that require reliable biometric identifications.

B. Poor Environment Conditions

Apart from that, the poor environment conditions such as lighting or heavy raining may also lead to failure to enrol biometric data using the facial recognition system. This is because overexposure or underexposure of light may wash out the details of the individual's face, making the system challenging to capture the accurate facial features of an individual.

Solutions

To address these challenges, high quality biometric scanners can be used in order to overcome the interference from moisture or dirt, and the biometric scanners or cameras should be cleaned consistently. Organisations, such as banks should allow alternative enrolment methods, which not only allows biometric authentication such as fingerprints or face recognition. It can also include the iris recognition or voice recognition for those individuals which have difficulties in scanning their fingerprints. Besides that, adaptive algorithms should also be implemented in the biometric authentication system especially for the face recognition not only to improve the accuracy of the captured biometric information but also allows the biometric authentication system to adapt to different environment conditions.

2. False Acceptance and Rejection Rates

Besides that, the false acceptance and rejection rates of a biometric system is also another challenge when implementing a biometric authentication system. The biometric system usually makes two types of mistakes, one of them is the “false positive” statement where the system incorrectly matches the input to the non-matching template or a “false negative” statement, in which the systems are unable to detect a match between the input and a matching template. The two statements above will lead to a probability of false acceptance and the rate of rejection will be increased in a system that is implemented with biometric authentications. This means that biometric systems will sometimes misidentify the users or individuals, which leads to unauthorised access or denied access to legitimate users or individuals. For example, identical twins may have similar biometric characteristics such as facial biometric. It is difficult for biometric technologies to detect and differentiate based on their faces. An identical twin sister is able to unlock her sister’s iPhone if her sister is using the IOS version that is earlier than 17.1 using the face recognition system that is implemented in iPhone. Moreover, the user interaction with the sensor may also differ between the enrollment and the recognition stages and may also lead to false acceptance, such as the pose or the angles when capturing the biometric information. An individual who got injured or has medical conditions may also result in changing a person’s biometric characteristics between the enrollment and recognition stages which will cause the rejection or failure of the authentication.

Solutions

As for the solution in order to overcome the challenges to implement biometric authentication as well as reduced the risk of identity theft, the organisations such as banks or systems can implement a hybrid systems which combines biometric modalities such as it needs fingerprint recognition and iris recognitions at the concurrent time to improve the accuracy of the biometric authentication systems. The users of a biometric authentication should also be educated or instructed on how to present their biometric properly so the biometric authentication system is able to capture and collect biometric data accurately and minimise the rejection rates.

3. The Cost and Resource Constraint

Additionally, another difficulty in implementing biometric authentication that should not be missed is cost and resource constraint. Biometric authentication systems have a high implementation cost because they need advanced and high quality hardware equipment in order to capture biometric information in great detail such as the fingerprint scanner, face recognition cameras and the iris scanners. Besides that, setting up biometric systems with high security measures can be expensive because it needs to be secure and robust to ensure that the biometric information will not be stolen by any unauthorised users. Therefore, implementation of biometric authentication systems can increase the manufacturing cost of an organisation. Moreover, maintenance of biometric authentication systems must be done and the frequency of doing so depends on the system itself to ensure that the systems are up to date which will further increase the cost. Operational cost of an organisation is increased as the biometric system needs to be constantly monitored, maintained in order for it to function properly and in a good condition. Furthermore, staff also have to be trained well enough in order for them to be able to know how to use the system in a proper manner which in the end, increases the overall cost of implementing biometric authentication systems.

Solutions

These challenges can be solved if the organisation implements the biometric system by phases or stages. This is because phased implementations of a biometric authentication system are able to spread cost over time as it can pay after every phase is implemented. Besides that, cloud-based biometric systems can also be considered as one of the methods to reduce hardware expenses as well as easier facility updates especially for those small to medium organisations. In addition, having a thorough review on the entire system once a year with the help of an expert would also cut costs for the organisation. By engaging with a seasoned expert such as a biometric consultant, the expert can provide valuable insights into system performance and recommend enhancements. This would help to ensure that the biometric system is up-to-date with the latest technology and standard while at the same time reducing the cost of the entire operation.

4. Legacy Systems

Integration of a biometric authentication system into an existing system of an organisation can also be challenging. This is because most of the organisation's existing systems are still using the older version of the operating system which might be not compatible with newer technology. These systems might be using outdated software and hardware, making it hard to modify and upgrade. The lack of flexibility of legacy systems not only increases the costs, it will also increase the project timelines to ensure that the biometric technologies are able to integrate and function well. Besides that, Integration process may be difficult and complicated as different biometric authentication systems may adhere to various standards or formats. Moreover, some departments in an organisation may have independent and incompatible systems which leads to inefficiency of data management and will also increase the difficulties of integrating the organisation's system and the biometric technologies.

Solutions

In order to overcome these difficulties, industry-wide standards should be proposed to streamline the implementation of biometric authentication systems as it can simplify the integration process and it is easier for organisations to adapt to new technologies without rebuilding their existing infrastructure. Another alternative that should be considered is that the users should upgrade their hardwares and softwares in order for them to be compatible with the biometric authentication system.

5. System Security

Last but not least, security of the biometric data is also one of the challenges of implementing biometric authentication systems. This is because biometric data is a high value target for attackers or hackers due to its usefulness and immutability after it has been compromised to unauthorised users. If the biometric data of an individual has been compromised, it could be used in any way such as identity theft, fraudulent transactions, unlawful access, social engineering attacks, biometric spoofing and others that would lead to possible catastrophic issues to the entire nation.

Solutions

Organisations need to have a secure and reliable security system in order to prevent the biometric information being data breaches. If unauthorised users are able to gain access to the biometric information, they are able to misuse it to do some illegitimate activities such as bank transactions without the notices of the bank accounts owner. Transport Layer Security (TLS) should also be implemented to ensure the biometric data is safe and secure from collecting point to the database systems to prevent data stolen in the middle of transferring or transmitting data. In order to ensure the safety of the biometric data, it should be saved and stored in an encrypted format, where the data remains unreadable without the decryption key or password.

1.4 Conclusion

To sum it all up, the implementation of the biometric authentication systems has made tremendous changes and significant improvements in terms of efficiency and security across multiple sectors.

Biometric authentication systems use unique biological traits such as fingerprints, facial recognition, voice recognition and iris recognition to verify users identity with accuracy. By doing so, it has played an important role in both safeguarding sensitive information and being one of the most convenient tools.

This can be proven by the sheer amount of usage of biometric authentication technology by typically everyone in our country. It has been very beneficial as it also speeds-up the efficiency of the verification processes while also making sure that the authentication process is the safest as it can be.

Different organisations and big companies have taken up the opportunity to apply these systems into their working environment, improving not only the security of their operations but also enhancing overall productivity. As the case studies above demonstrate, other than the fact that the implementing biometric authentication systems have set a new standard to the security industry, it has also changed the way various sectors approach identity verification and access control, influencing the industries beyond just security, including finance, healthcare, and retail, where the need for reliable and efficient authentication methods is paramount. Moreover, this technology also notably minimised a lot of the risks that traditional security threats used to pose. Due to the positive review and the clear impacts the biometric authentication system had made, it raised the bar for security protocols, pushing organisations to adopt more advanced and user-friendly technologies that enhance both protection and convenience.

As we look beyond the future of biometric authentication systems, we should consider the key areas that will shape their continued evolution and impact on computer security. Hence, there are some suggestions that we should consider as guides for the future directions of implementing a biometric authentication system. Suggestions such as below:

1. Multimodal Biometric Integration

The usage of multiple biometric indicators by personal identification systems for identifying the individuals. This would benefit individuals especially those whose biometric may be compromised due to certain conditions by combining different biometric modalities to reduce the likelihood of errors or false rejections.

2. Artificial Intelligence (AI) and Machine Learning

AI and machine learning can improve the accuracy and adaptability of biometric systems by detecting spoofing attempts and adapting to changes in biometric traits over time. We should also focus on utilising machine learning algorithms which can continuously learn the patterns of the new potential security threats that are more sophisticated and come up with solutions to combat these threats.

3. Challenges of Implementing Biometric Authentication Systems in Internet of Things (IoT)

Implementing biometric authentication systems in IoT should be one of the primary future directions to be focused on. The main reason for doing so is it helps to maintain data integrity by securing sensitive user data and preventing security breaches and data theft. With that being said, one of the primary challenges is resource limitation of many IoT devices. IoT devices often have constrained processing power, memory and battery life which will make it difficult to run complex algorithms required for biometric authentication. It is crucial that we address the challenges such as the above mentioned to realise its full benefits without compromising performance, privacy, or user trust.

4. Strengthening the System's Resilience Against Sophisticated Attacks

Attacks such as Deepfake attack occur when an individual's appearance or voice is altered to look like someone else in order to carry out the plan that the attacker has in mind. This is an example of a new security threat that is made to counter biometric identification specifically. The clear suggestion would be to carry out research and investigation in order to improve the security of the biometric authentication system even further to a point where the occurrence of said attacks is near zero.

5. Ethical and Legal Considerations

As the use of biometric authentication systems becomes increasingly prevalent, it's important that we should address the ethical and legal concerns that follow along with this technology as questions about privacy, data protection, and the potential for misuse of biometric data has become a popular topic among the discussions of the citizens. The ethical implications of biometric authentication should be throughout explored while at the same time regulations and standards that govern the use of biometric data should be created to act as a guideline during the implementation of biometric authentication system

2.0 Overview

To be concise, Biometric authentication systems have become a foundation of modern computer security, leveraging unique physical and behavioural traits of an individual such as fingerprints, facial recognition, iris recognition and voice patterns to verify identities with greater accuracy and absolute convenience. As given some real world applications and case studies of biometric authentication systems above, it is clear as day to say that these systems offer significant advantages over traditional computer security methods such as passwords and pin number while also bypass the majority of the vulnerabilities that are addressed when using said traditional methods, providing a higher level of security while simplifying user access. Their widespread usage across various sectors highlights their effectiveness in safeguarding sensitive information and combating threats.

To talk about the broader context of computer security, biometric authentication plays a vital role in addressing the increasing sophistication of cyber threats as that is what it is made for. The fact that before the era in which biometric authentication made its way to the industry, a lot of the vulnerabilities of traditional methods of computer security have been found out and exploited by attackers and threat actors. Countless ways and methods have been created by attackers to bypass security and gain unauthorised access to literally everything with just little effort in doing so. Moreover, as time passes, more and more attackers are starting to appear everywhere as they march toward the same goal which is to satisfy their own needs and desires without having a thought whether their actions are deemed immoral and illegal. Hence, biometric authentication comes into play by making it more difficult for unauthorised users to gain access. These systems help protect against identity theft, data breaches, and other forms of cybercrime. However, as their use expands, it is crucial to also consider the ethical and legal implications, ensuring that these powerful tools are implemented responsibly and in ways that respect privacy and human rights.

2.1 Threat Analysis

Without the usage of biometric authentication systems, traditional authentication methods, such as passwords and PINs are often vulnerable to a number of attacks such as identity and credential theft, spoofing, insider threats, social engineering attacks and so on. These attacks are brutal as these attacks have ruined innocent lives while satisfying the attackers wishes and desires. Below are the example threats and attacks that are famous and can be seen everywhere.

Identity Theft

Identity theft happens when someone steals and uses another person's personal information to commit fraud and other crimes. Sensitive information such as social security numbers, addresses and other information can be used by the perpetrator to open new accounts and make outlandish purchases or to affect the victim's credit or financial standing. Identity theft can be easily committed due to the fact that traditional security methods leave personal data more exposed to potential theft. This would cause long-term consequences such as damaged credit scores, legal complications, and emotional distress for the victim.

Real Life Example

Based on (*U.S. Announces What Is Believed The Largest Identity Theft Case In American History; Losses Are In The Millions (November 25, 2002)*, n.d.). To show a real life example, in 1999, a perpetrator known as Phillip Cummings who was a help desk worker at a software firm, took a spreadsheet of logins and passwords when he quit, granting him access to a number of credit reports. He took the reports and sold it to a group of criminals which each of them worth from \$50 to \$100 million. This massive identity theft scheme spanned nearly three years and involved more than 30,000 victims. To date, more than \$2.7 million in financial loss has been confirmed.

Credential Theft

Credential theft happens when someone steals and uses another person's specific login details such as usernames, passwords, or PINs. These stolen credentials are often used to gain unauthorised access to any accounts such as personal, financial, or corporate accounts in order to carry out malicious activities. These activities include transferring funds, accessing confidential information, launching further attacks within a network, or even impersonating the victim to perpetrate fraud or other criminal acts. Credential theft would lead to massive consequences such

as financial losses, data breaches, loss of intellectual property and significant reputational damage.

Both identity theft and credential theft occur right after attacks such as below

Spoofing

Spoofing attack is one of the most famous attacks as it is difficult to prevent due to its deceptive nature. Victims are usually unaware that they are interacting with a malicious entity until it is too late. Spoofing attack occurs when someone or something disguises themselves to be something else that the victims believe in an attempt to gain a victim's confidence, get access to a system, steal data, or spread malware. The attackers can disguise themselves as a trusted individual, organisation, or website with the aim of deceiving the victim. This kind of attack would bring a lot of consequences such as sensitive information being stolen and misuse or malware installation on victims' devices without their consent which will compromise the victim's system and network

Spoofing attacks come in many forms. One of them being email spoofing where the attacker appears to be a trusted source by manipulating the email header. This would allow them to appear to be someone or something that the victims would not doubt and will put their trust on them without a second thought. Spoofing usually goes hand in hand with phishing, which is another form of attack where the attacker deceives the victim into revealing sensitive information or clicking a suspicious link. For example, the spoofed email may contain a link to a phishing designed website to steal personal information and login credentials. It may also contains malware attachment that can compromise the victim's system and network once downloaded

Real Life Example

Based on (US Department of Justice, n.d.), here is a real life example of phishing email: the attackers worked their way around the Clinton campaign's top-of-the-line digital security in order to steal chairman John Podesta's emails during the 2016 U.S. presidential election. John Podesta fell victim to the email phishing scam. He was deceived by the attackers' email that appeared to be a security alert from Google to change his password. Since he thought the email to be legitimate, he clicked the link that was in the content of the email and entered his credentials on a fake website, allowing the attackers to gain access to his emails. The stolen emails were later published which led to significant political fallout and becoming a major controversy during the campaign.

Brute Force Attack

As the name suggests, a brute force attack is a method where the attacker uses trial and error to crack passwords, login credentials and encryption key in order to gain access to a specific account that the attacker targeted. This method of attack is simple yet reliable as the attacker can use automated tools that are highly advanced enough to test thousands or even millions of combinations in a short period of time to crack open an account. The few downside to this for the attacker is that it can be time consuming depending on the complexity of the password that was set for the account. Brute force attack also requires significant computational power but it is simple to execute as it does not require the attacker to have some sort of hacking skills. A successful brute force attack attempt will cause reputational damage, the possibility of your personal information and credentials being compromised and sold to a third party, being put in a situation where you are being threatened and demanded to pay a ransom to the perpetrator and so much more.

Real Life Example

Based on (Seals, 2016), here is a real life example of a successful brute force attack. In 2016, The popular eCommerce platform, Alibaba became a victim of a successful brute force attack that compromised around 20.6 million users' accounts by using a database of 99 million usernames and passwords. Experts revealed that the main cause for the attack to be successful was the overlapping of passwords by users. It was discovered that the majority of the users were using the same credentials and passwords on the platform for their other accounts. Some users also used weak passwords that are incredibly easy to be figured out. Due to the negligence of the company in computer security, it has caused one of the biggest brute force attacks known to man.

Deepfake

Deepfake attack is an attack that involves the use of deep learning algorithms to create fake images, videos, or audio recordings that are highly realistic. These synthetic media are very convincing as they mimic the appearance, voice or behaviour of real people since they are designed to deceive anyone to believe that they are genuine. Based on (*What Is a Neural Network? - Artificial Neural Network Explained*, n.d.), the technology that deepfake relies on is neural networks which is a method in artificial intelligence that teaches computers to process data in a way that is inspired by the human brain. Deepfake specifically uses Generative Adversarial Networks to generate or alter media content, creating lifelike simulations by analysing vast amounts of data, such as images or recordings of the target individual which is easily obtained either from the internet through social media or by other means like an insider source. Deepfake can lead to several harms such as loss of employment opportunities, public humiliation, or damage to personal relationships.

Real Life Example

A real life example that uses deepfake for malicious purposes was South Korea's Digital Sex Crime. South Korea has a dark history about the abuse of artificial intelligence technology to make pornographic deepfake videos using real life victims in their teens. Based on Writer, S. (2024, September 2). South Korea faces AI-driven proliferation of deepfake sex crimes. Nikkei Asia, A 28-year-old man was sentenced to five years in prison for producing sexually explicit deepfakes using photos of women including alumni of Seoul National University. Another four men were charged in relation to this case where these perpetrators used images of dozens of women from the university through the victim's yearbooks or social media profiles to create and distribute at least 400 of deepfake videos. They used Telegram as a means of distribution. Channels for this purpose had been increasing rapidly on the platform, often sorted based on the victim's ages and occupation. One of the channels reportedly amassed more than 220,000 users. The following case has not stopped as a new case arises which is the "nth room" case. Based on (*Nth Room Case - Wikipedia*, n.d.), this case involves underage girls who were coerced into sending sexually explicit videos that were circulated online. These crimes have significantly contributed in increasing the number of deepfake-related sexual crimes from January to July of 2024 which nearly double the total for all of 2021. Not only did deepfake started a new series of crimes but also making these crimes a long-term problem for developed country such as South Korea.

Replay Attacks

Based on (*A Guide to Replay Attacks And How to Defend Against Them*, 2024), replay attack is a form of network attack where an attacker intercepts and retransmits data that was previously exchanged between two parties. Replay attacks can capture various forms of authentication data such as passwords, session tokens, or cryptographic authentication hashes. To lay out an example, a company's staff member requests for a financial transfer by sending an encrypted message to the company's financial administrator. The attacker can take up the chance to eavesdrop in this message and capture all the sensitive information such as credentials. The captured information can be used later to grant the attacker access to the victim's financial account and make any amount of financial transfer to any account of the attacker's choice at any given time. The worst part about it is that the victim of such cases does not have any means to know such acts until they realise they went bankrupt in a blink of an eye.

Real Life Example

Based on (*Replay Attack - Wikipedia*, n.d.), an example that can be mentioned is the replay attacks involving automotive keyless entry systems. In 2022, vulnerability in the Remote Keyless Entry (RKE) Systems of certain vehicles was exploited. The attackers perform a rollback replay attack which is a new replay-and-resynchronize attack against most of today's RKE systems. This attack uses a device that can receive and transmit radio waves within range of the target vehicle. The attacker uses the device to jam any radio frequency signal that unlocks the target vehicle while receiving it and placing it in a buffer. The buffer can be reuse by the attacker to later replay the signal in order to unlock the vehicle when the owner is not around with the remote key

Insider Threats

Insider threat refers to a risk posed to an organisation or a system by an individual who has inside access to the organisation's entire data. Information and facilities depending on the level of their authority and access. Individuals such as employees, contractors, business partners, or vendors who have malicious intent or agenda have been given legitimate access to the organisation's system, allowing them to steal trade secrets, sabotage the system or commit fraud. This would cause the organisation to face data breaches which create a snowball effect leading to potential financial losses and reputational damage to the organisation.

Real Life Example

Based on (Snowden, n.d.), A famous case that can be used as a real life example is the incident involving a man named Edward Snowden, who is a former NSA contractor, leaked classified information about the U.S government's global surveillance program to journalists in documents. His goal was to inform the public about the clear extent of government surveillance and to spark a debate on privacy and civil liberties. The leaked documents have started public outcry and the U.S. government faced legal and political repercussions. Edward Snowden was viewed as a hero who exposed the government overreach by some citizens of the United State while some deemed him as a traitor who endangered national security. Even though this is a positive outcome of a case of insider threat, it is still a reminder of the potential dangers posed by insider threats with access to sensitive information and having the skill of an expert in the field.

The threats and attacks mentioned above are addressed by the technology known as biometric authentication systems due to the fact that their very existence has significantly raised the bar for security protocols. In the next topic we will talk about how these systems addressed the issues mentioned and how much of an impact it has brought to the field of computer security.

2.2 Effectiveness Evaluation

The new technology, biometric authentication system has taken the world by storm in the field of computer security due to the fact that it has significantly contributed in enhancing security in a big way. The reason is that their reliance on unique physical or behavioural characteristics of an individual such as fingerprints, voice recognition and the like has effectively mitigated a ton of threats that have been haunting innocent people for a long time. Yet, how much effectiveness these systems have to offer depends on a range of factors, which includes their implementation, vulnerabilities and user adoption.

Password's Replacement

One of the first things to talk about regarding the effectiveness of biometric authentication systems in security is the fact that it has replaced one of the most traditional security methods, which is password-based authentication. It cannot be denied that biometric authentication is much safer than using password-based authentication. For example, Biometrics such as fingerprints are unique to every single individual, this proves the fact that biometrics are virtually impossible to forge or steal. In addition to being more secure, it is to be said that the level of convenience that password-based systems provide is clearly outmatched by biometric authentication systems. This is due to the fact that passwords can be easily forgotten and the same combination of a password can be reused by anyone across multiple platforms which will very likely increase the probability of the occurrence of security breaches.

Based on (*The United States of P@ssw0rd\$, n.d.*), A survey was conducted on a nationally representative sample of 3419 adults living in the U.S. 24% of adult Americans have used common passwords and 59% of adult Americans have incorporated a name or a birthday into their password to an online account. 66% of adult Americans use the same password for more than one online account. This proves that passwords nowadays are very vulnerable to attacks such as brute force attacks. With the replacement for password, biometric authentication covers a lot of the vulnerabilities that passwords have.

Threats Mitigated By Biometric Authentication

1. Identity Theft, Credential Theft, Spoofing Attack & Brute-Force Attack

Unlike passwords, which can be shared, stolen or guessed, biometric authentication systems use unique physical or behavioural traits which are tied to individuals. Biometric data is difficult to forge or steal thus mitigating the risks of unauthorised access via identity theft, credential theft, spoofing attacks, which the attacker pretends to be someone else using the individual's credentials to trick the victims and brute-force attacks which the attacker relies on mathematical relationship of the password that ties with the victim who could be using a password with weak combination.

2. Phishing and Social Engineering attack

Password-based systems are defenceless against phishing and social engineering attacks. The attackers trick users into revealing their passwords through emails or any links to a suspicious website that the victims accidentally clicked. Biometric authentication systems greatly mitigate this risk as it is nearly impossible to "trick" someone's biometric data such as fingerprint into granting full access. Unlike password-based systems, biometric authentication systems do not rely heavily on the user's ability to remember and protect their own credentials, making it more arduous for the attackers to act in gaining unauthorised access.

3. Replay Attack

The attacker performs a replay attack by intercepting and retransmits data that was previously exchanged between two parties. Basically, the attacker got a copy of the victim's password and can be used repeatedly if needed in order to have unauthorised access. The fact that biometric authentication relies on the real-time capture and verification of biometric data which is inherently dynamix, making it harder to be replicated.

Dynamic data is information that is periodically updated, it will keep changing as new information becomes available. An example would be fingerprints, where every time an individual tries to scan their fingerprint to unlock their smartphones, they could be in different locations, different times of the day, which will highly affect the outcome of the biometric input for every scanning attempt. The variability of biometric data makes replay attacks difficult to perform successfully as the exact same biometric input cannot be captured and used repeatedly.

4. Insider Threat

Insider threat actors mostly gain unauthorised access by being a trusted individual of a company and were given access to the organisation's system. In the past, most organisations and companies used passwords and other credentials of an individual of higher authority in the organisation to secure their systems. These passwords are easily stolen by insider threat actors. With the implementation of biometric authentication into the systems of organisations and companies, the threat posed by insider threat actors is significantly reduced due to the fact that biometric data is unique to each individual and cannot be transferred between users. This helps to ensure that access to critical information of the organisation is tied directly to an individual's physical or behavioural characteristics, making it difficult for credentials to even have a chance to be misused by an insider.

Vulnerabilities of Biometric Authentication System

Biometric authentication has offered significant advantages in enhancing security, but it still has some downsides and vulnerabilities that should be considered and addressed.

1. False Acceptance and False Rejection

Based on (*Everything About FAR and FRR*, n.d.), some biometric authentication systems are not perfect due to the complexity of the technology. Some systems have False Acceptance Rates, where unauthorised users are mistakenly granted access, or False Rejection Rates, where legitimate users are denied access. These two go hand in hand in determining the system's security level and ease of use. When aiming at lowering the False Acceptance Rate, this would increase the system's security level while lowering the ease of use for users. If it's the other way around, aiming at lowering the False Rejection Rate, this would increase the ease of use for the users while lowering the security level at the same time.

Only a few systems facilitate both a high level of security and user friendly access system. If organisations are only given two choices either using the system with higher security level or higher ease of use and they can only pick one, most of them will likely pick higher ease of use at the expense of lower security. This becomes a problem as it creates a false sense of security for the users since the user is not aware of how the software and the system are set up.

2. Not 100% “Threat-proof”

Biometric authentication has covered most of the threats by mitigating them until a degree where they no longer pose a threat. However, there are still a few threats that are not immuned by the system. That is because the lack of advancement made on biometric security and also the sophistication of the threats keep on increasing as time passes. An example of a threat that can counter biometric authentication systems is deepfake attack, which counters systems that especially rely on facial recognition and voice recognition. Based on (Murphy, 2024), some Deepfake technologies that are highly sophisticated are able to use artificial intelligence to create highly realistic, synthetic versions of an individual’s face or voice and it can be used by the attacker to deceive the biometric authentication system, bypassing security in order to gain unauthorised access

Other Implications

1. Cost of Implementation

Biometric authentication are expensive to implement and maintain due to the need of specific hardwares such as fingerprint scanners, facial recognition cameras etc. and softwares that allows the devices to operate together with these hardwares

2. Legal and Ethical Implications

There are issues regarding data ownership, consent and also misuse become significant concerns of the public. Every organisation around the world that uses biometric technology, including organisations that have ties with the government must comply with regulations to ensure biometric data is safe

2.3 Future Directions

It is important that future directions should be thoroughly discussed in order to make clear decisions that could drastically improve the level of security and ease of use of the biometric authentication system until it no longer has a shred of vulnerabilities that could be exploited and issues that could potentially shake the very core of computer security.

Multimodal Biometric System

One of the main future directions that should be considered first is to implement multimodal biometric into all systems in effort to increase the level of security. Based on (Ughade, n.d.), multimodal biometric is the combination of multiple biometric traits such as fingerprint, facial recognition, iris recognition and so on in a single system for enhanced security and accuracy. This would address the limitations of relying on a single biometric trait to identify an individual's identity. Cases where some biometric traits can be hard to capture in some conditions that are out of human control.

Integration of Artificial Intelligence and Machine Learning Models

Based on (Yurdasen, 2023), The integration of Artificial Intelligence (AI) and machine learning models is a great suggestion as a future direction that developers of biometric authentication systems can follow. This could help to improve the accuracy of the system and also its ability to adapt to any anomalies that it encounters. AI and machine learning models can help to detect any sort of threats and attacks that had occurred before in the past as well as potential new threats and attacks. In terms of adaptability, it can help biometric authentication systems to adapt to different inputs of biometric data that are scanned in different conditions.

Combating Sophisticated Threats and Attacks

Attentive research and investigation should be carried out by the developers of biometric authentication systems is another future direction that should be taken as the level of danger of the threats and attacks is immeasurable and ceaseless. It is crucial to constantly make improvements and adjustments to the biometric authentication system as more and more sophisticated attacks and threats are appearing left and right. An example is enhancing liveness detection techniques to a point that it can detect and distinguish between real users and deepfake technologies that are used for impersonation.

Ethical and Legal Challenges

Global rules and regulations should be created and used to standardise the collection, storage management and the use of biometric data at a global level. This could help to ensure that biometric information is strictly protected in terms of privacy. Biometric data should be collected only when necessary and it should be encrypted and stored securely. In addition, biometric data should not be shared to any third party for any reason other than authentication purposes.

2.4 Conclusion

In conclusion, biometric authentication becomes a critical advancement in the broader landscape of computer security technologies by leveraging unique human physical and behavioural characteristics which have elevated the standards of cybersecurity. The main highlight of biometric authentication is its importance in addressing various threats which traditional security methods are not able to overcome. Unlike passwords, biometric data cannot be easily forged or lost by the users.

Even so, there are still issues such as privacy concerns as well as new potential sophisticated threats like deepfake attacks should not be left unattended for as these challenges demonstrate no security method is entirely invincible. Moreover, in order to protect individual privacy and to prevent misuse of data, ethical and legal implications bound to the collection and storage management of biometric data must be investigated and examined attentively.

All in all, even while biometric authentication is not 100% threat-proof, It still becomes one of the biggest contributors to the overall security and the robustness of modern computer systems by mitigating cyber threats to ensure everyone's safety.

References

1. <https://blog.hidglobal.com/2024/01/10-biometric-trends-watch-2024#:~:text=Here%20are%2010%20key%20trends.scans%20and%20even%20gait%20recognition>. (*10 Biometric Trends to Watch in 2024*, 2024)
2. <https://www.thalesgroup.com/en/markets/digital-identity-and-security/government/inspired/history-of-biometric-authentication#:~:text=The%20History%20of%20Biometric%20Authentication&text=Biometrics%20can%20be%20traced%20back,the%20work%20of%20Alphonse%20Bertillon> (*Biometry Authentication History (Infographic)*, 2023)
3. <https://optimalidm.com/resources/blog/types-of-biometrics-sensors/> (*Types Of Biometric Authentication | What Is Biometrics?*, n.d.)
4. <https://www.signicat.com/blog/face-recognition#:~:text=By%20definition%2C%20facial%20recognition%20refers,works%20like%20a%20face%20scanner>. (Zaragoza, 2024)
5. <https://www.nec.com/en/global/solutions/biometrics/iris/index.html> (*Iris Recognition: Biometric Authentication | NEC*, n.d.)
6. <https://www.innovatrics.com/fingerprint-technology/#:~:text=Fingerprint%20recognition%20is%20the%20process,series%20of%20ridges%20and%20grooves>. (*Fingerprint Technology*, n.d.)
7. <https://support.apple.com/en-my/guide/security/sec067eb0c9e/web> (*Face ID and Touch ID Security – Apple Support (UK)*, 2021)
8. <https://support.apple.com/en-us/105095#:~:text=On%20iPhone%20and%20iPad%2C%20a,subepidermal%20layers%20of%20your%20skin> (*About Touch ID Advanced Security Technology*, 2023)
9. <https://www.hsbc.com.my/ways-to-bank/telebanking/#started> (*HSBC Telebanking | Benefits and How to Get Started - HSBC MY*, n.d.)
10. https://www.researchgate.net/publication/268334914_Security_Issues_on_Identity_Card_in_Malaysia ((PDF) *Security Issues on Identity Card in Malaysia*, n.d.)
11. <https://ovic.vic.gov.au/privacy/resources-for-organisations/biometrics-and-privacy-issues-and-challenges/> (*Biometrics and Privacy – Issues and Challenges – Office of the Victorian Information Commissioner*, n.d.)

12. <https://identitymanagementinstitute.org/biometric-authentication-challenges/> (*Biometric Authentication Challenges*, 2021)
13. <https://alicebiometrics.com/en/biometric-authentication/> (*Biometric Authentication: Cost and Benefit Analysis*, n.d.)
14. <https://hyperverge.co/blog/future-of-biometrics/> (Ughade, n.d.)
15. <https://www.techtarget.com/searchcustomerexperience/definition/voice-recognition-speaker-recognition> (Gillis & Mellon, n.d.)
16. <https://www.justice.gov/archive/criminal/cybercrime/press-releases/2002/cummingsIndict.htm> (*U.S. Announces What Is Believed The Largest Identity Theft Case In American History; Losses Are In The Millions (November 25, 2002)*, n.d.)
17. https://en.wikipedia.org/wiki/Podesta_emails (US Department of Justice, n.d.)
18. <https://www.infosecurity-magazine.com/news/massive-bruteforce-attack-on/> (Seals, 2016)
19. <https://aws.amazon.com/what-is/neural-network/#:~:text=A%20neural%20network%20is%20a,that%20resembles%20the%20human%20brain> (*What Is a Neural Network? - Artificial Neural Network Explained*, n.d.)
20. https://en.wikipedia.org/wiki/Nth_Room_case (*Nth Room Case - Wikipedia*, n.d.)
21. <https://asia.nikkei.com/Spotlight/Society/Crime/South-Korea-faces-AI-driven-proliferation-of-deepfake-sex-crimes#:~:text=The%20National%20Police%20Agency%20has,creating%20deepfake%20videos%20with%20AI> (Writer, S. (2024, September 2). South Korea faces AI-driven proliferation of deepfake sex crimes. Nikkei Asia)
22. <https://www.packetlabs.net/posts/a-guide-to-replay-attacks-and-how-to-defend-against-them/#:~:text=Replay%20attacks%20are%20a%20form,in%20Dtransit%20in%20cleartext%20form> (*A Guide to Replay Attacks And How to Defend Against Them*, 2024)
23. https://en.wikipedia.org/wiki/Replay_attack (*Replay Attack - Wikipedia*, n.d.)
24. https://en.wikipedia.org/wiki/Edward_Snowden (Snowden, n.d.)
25. <https://storage.googleapis.com/gweb-uniblog-publish-prod/documents/PasswordCheckup-HarrisPoll-InfographicFINAL.pdf> (*The United States of P@ssw0rd\$, n.d.)*
26. <https://recogtech.com/en/insights-en/far-and-fr-security-level-versus-ease-of-use/> (*Everything About FAR and FRR*, n.d.)

27. <https://www.techtarget.com/searchsecurity/tip/How-deepfakes-threaten-biometric-security-controls> (Murphy, 2024)
28. <https://www.aratek.co/news/how-artificial-intelligence-ai-is-used-in-biometrics> (Yurdasen, 2023)